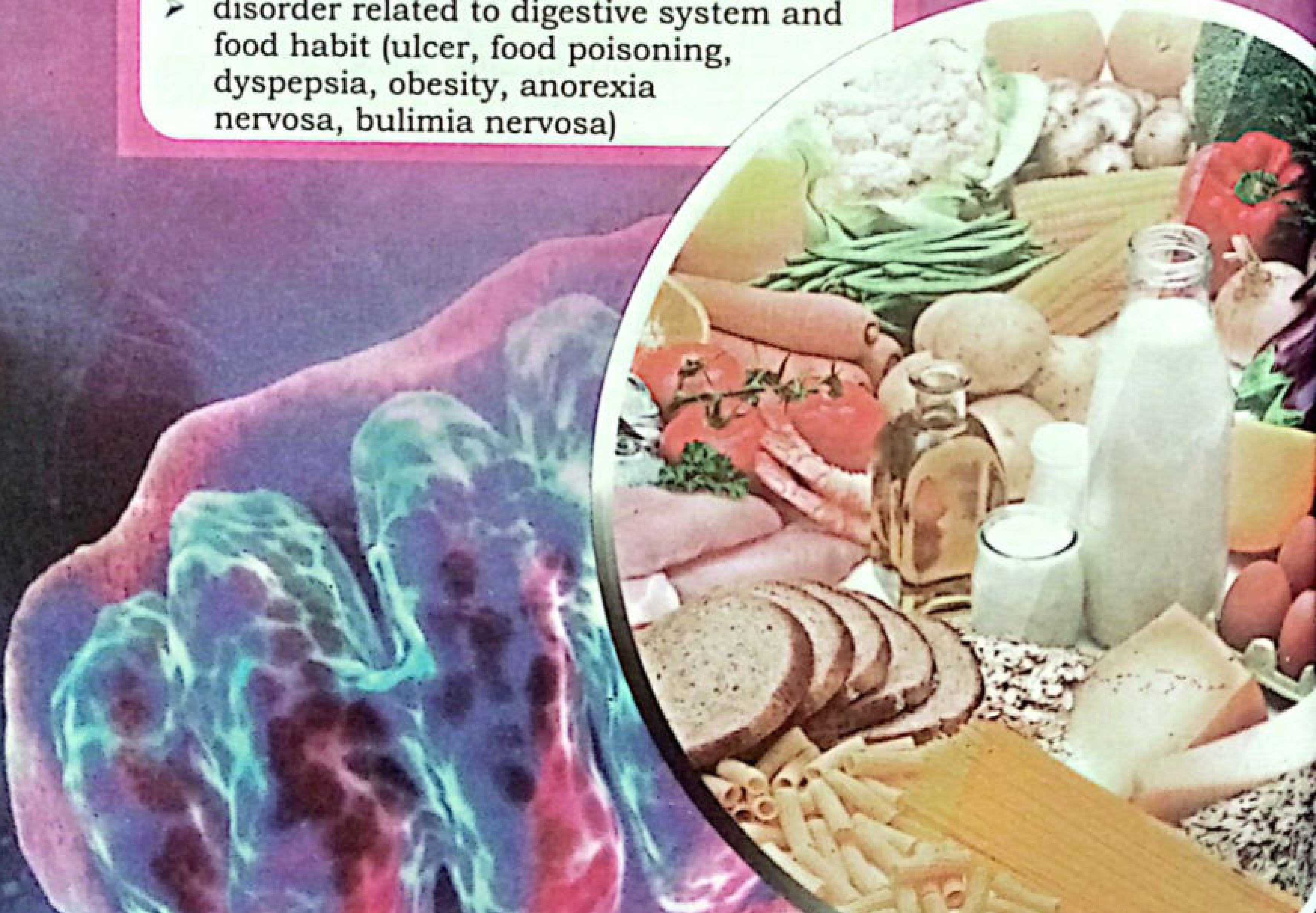
HOLOZOIC NUTRITION

Chapter 11

Major Concept

In this Unit you will learn:

- Holozoic nutrition
- Digestive system of man
- Alimentary canal; structural and functional details.
- Role of accessory glands
- disorder related to digestive system and food habit (ulcer, food poisoning, dyspepsia, obesity, anorexia nervosa, bulimia nervosa)



11.1.1 Holozoic nutrition

Just like a vehicle needs fuel to move, a living cell needs supply of **food** to perform its various biological functions. Through this process, the cells not only obtain energy but also get materials to grow and repair themselves. The food of an organism consists of different substances, the **nutrients** which are essentially required by the protoplasm to perform its different biological functions.

In the process of heterotrophic nutrition, an organism depends upon ready-made food like carbohydrates, proteins, etc., being prepared by other organisms, plants as well as animals. Holozoic (Gr. Holo=Whole; zoikos=of animals) nutrition is one of the type of heterotrophic nutrition which commonly occurs in animals. It is characterized by ingestion or taking in of food in solid or liquid form from the environment. It consists of ingestion, digestion, absorption, assimilation and egestion. Let's discuss them one by one.

Ingestion:

Taking in of food into the cell (in case of unicellular organisms) or body (in case of multicellular organisms). It takes place either through cell surface or through specific opening, **mouth.**

Digestion:

It is the process of breaking down of complex or non-diffusible food into simple or diffusible molecules so that they may be incorporated in the metabolism. This conversion process takes place chemically with the help of enzymes as well as mechanically.

Absorption:

Through the process of absorption, the soluble molecules of food are absorbed by the digestive membranes such as intestinal lining, etc., so that they may be utilized during the metabolic activities.

Assimilation:

The process of utilization of the absorbed molecules during the metabolic processes of cell is called assimilation.

Egestion:

The process of removal of undigested food from the cell/body is called egestion.

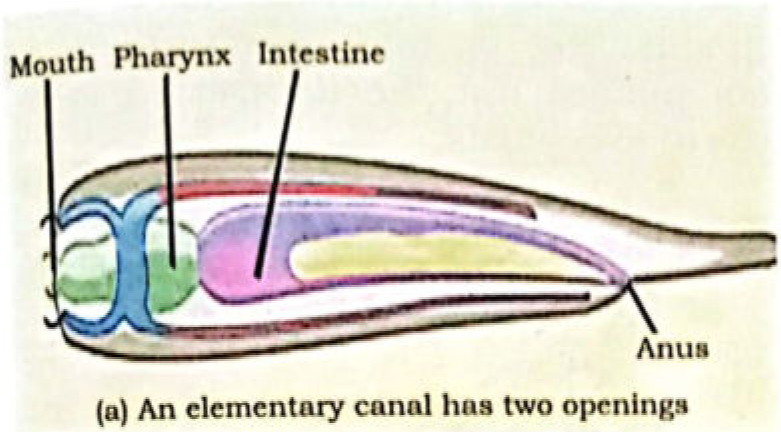
11.1.2 Intracellular and extracellular digestion:

Single cells organisms like protozoa commonly take in food particles present in their environment through their cell surface or plasma membrane. Thus the process of digestion takes place inside their cell. Such organisms do not release their digestive enzymes outside their cell. This kind of digestion is termed as **intracellular digestion**. Whereas on the other hand, majority of the multicellular animals, like round worms, insects, mammals, etc., take the food inside their body in a specific location called digestive tract/alimentary canal or gut made up of cells. They cannot take it inside their cells directly since the size of food particles is too large to be taken up by the cells. While the food remains outside the cells in their gut, the enzymes are released from the secretory cells into the gut where the food is being digested and converted into simple substances. The latter, is termed as **extracellular digestion**. Both types of digestions involve the process of the breaking down of food not only through **enzymatic conversion or chemical digestion** but in extracellular, also through some kind of mechanical conversion in many animals. The former is termed as **Chemical digestion** while the latter is **Mechanical digestion**. During the process of enzymatic break down of complex food, water is essentially required to facilitate the hydrolytic breaking down of food. On the other hand, mechanical digestion involves some kind of mechanically or physically breaking down (such as mastication, churning, grinding, peristalsis, etc.) of larger particles of food into smaller ones. It facilitates the process of chemical digestion thereby providing more particles of food to be acted upon by digestive juices.

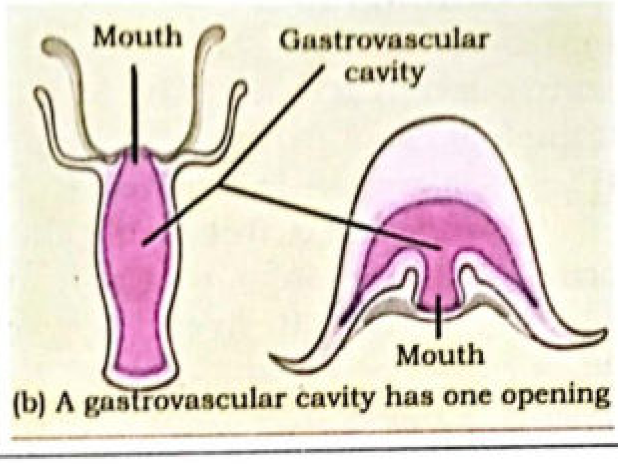
Sac-like and tube-like digestive system:

Tube like digestive system	Sac like digestive system
1. Animals have two openings in their digestive tract	1. Animals have single opening
2. Present in relatively advanced group of organisms	2. Present relatively in lower group of organisms
3. Food entrance and waste removal takes place by separate openings	3. Food entrance and waste removal takes place by single opening
4. Digestive system completes with associated glands and organs	4. Digestive system is incomplete and without associated glands and organs
5. Mechanical and chemical digestions both takes place actively	5. Mostly chemical digestion takes place actively
6. Digestive tract with variable pH due to complex nature of food	6. Digestive tract with not much variable pH due to simple nature of food
7. Enormous enzymes present which act upon food components	7. Fewer enzymes present which act upon food components
8. Digestive tract well supported by muscles	8. Digestive tract is not well

for strong peristalsis and defecation reflexes	supported by strong muscles



(a) An elementary canal has two openings



(b) A gastrovascular cavity has one opening

11.1.3 Digestion in *Amoeba*

Amoeba is a unicellular, eukaryotic organism which belongs to Protozoa. Being microphagous feeder, it feeds upon minute food particles and other microscopic organisms like bacteria, etc. The food is captured and ingested through its pseudopodia which are temporary finger-like extensions of cytoplasm. Pseudopodia are given out in the direction of food, form a cup-like structure or **food cup**. As soon as the food particles are trapped into the food cup, their tips start moving towards each other till completely enclosing the food. Thus the food is now enclosed in a sphere formed by the membrane which pinches off from the plasma membrane and moves into the cytoplasm. This tiny sphere of plasma membrane containing food particles and water is termed as **food vacuole**. The entire process is ingestion. The food vacuole thus moving inside the cytoplasm is attached with lysosomes containing different hydrolytic enzymes which act upon variable pH ranging from 5.6 to 7.3 depending upon the nature of food components and initiate the process of digestion. Since the food is digested inside the food vacuole, it is termed as **intracellular digestion**. The digestive vacuole moves deeper into the cytoplasm through the streaming movement of cytoplasm(cyclosis). After the digestion process is completed, the food vacuole gives out numerous fine, cytoplasmic canals. The diffusible food particles move into these canals for the absorption process. The absorbed food particles then diffuse out into the different parts of the cytoplasm where they may be assimilated accordingly. The digested food may be incorporated to form new protoplasm or it may be broken down to liberate energy. Meanwhile, in the food vacuole, there

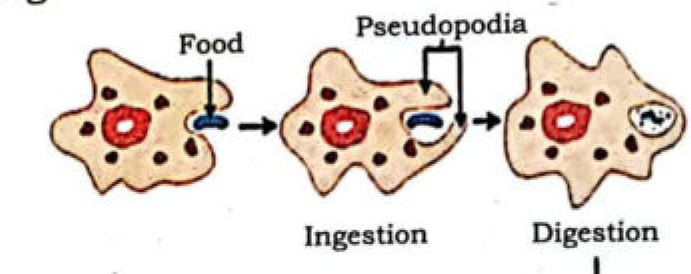


Fig: 11.1 Digestion in amoeba

is some undigested food left behind. The vacuole containing undigested food is pushed towards the cell surface where it attaches with interior of plasma membrane. At the point of attachment, the plasma membrane breaks towards exterior side. The cytoplasm pushes out the undigested food to completely egest it through the process of exocytosis.

11.1.4. Digestion in *Planaria*

Planaria is a free-living flat worm which belongs to Phylum Platyhelminthes. It lives in wet places under rocks, close to streams, canals and in brackish water. It has a soft, elongated, unsegmented and dorsoventrally flat body. The digestive system is very simple and considered to be **incomplete or sac-like**. The mouth lies on a small, muscular tube, **the pharynx** lying almost in the middle of the body. The pharynx is attached to intestine through a very short **esophagus**. The intestine divides into three blind branches, one anterior and two posterior-lateral ones. Each of the three branches, gives out numerous branches or **diverticula**. Each diverticulum further branches into very fine and blind branches or **ramification**. There is no anus.

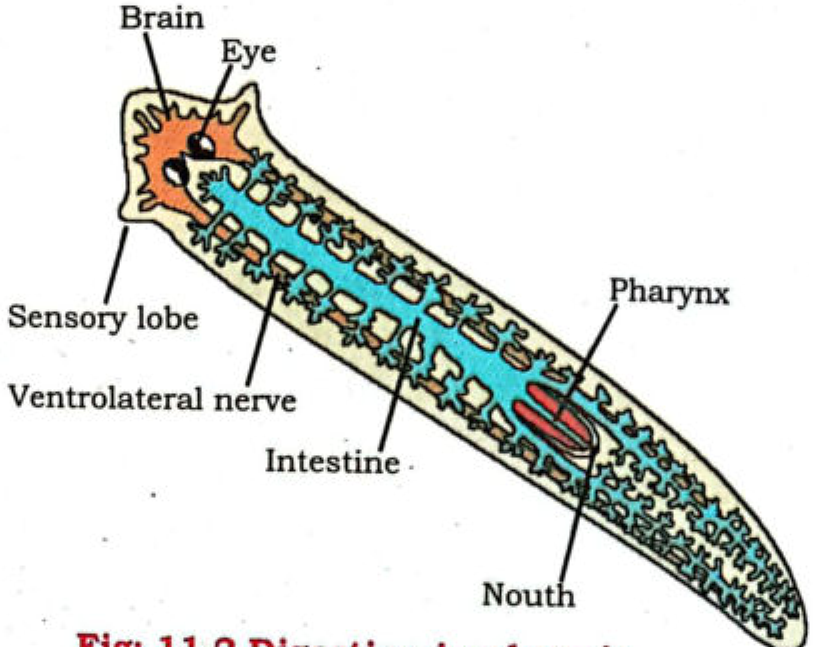


Fig: 11.2 Digestion in planaria

The planarians are carnivore. They capture small larvae, small insects, etc., through trapping them into thick mucussecreted from the pharynx. The prey is then sucked up inside the pharynx through the mouth. The glandular cells secrete variety of enzymes to begin the process of extracellular digestion from the pharynx till the end of intestine. The muscular action of the body pushes the food throughout the three branches of intestine. As the food breaks up into smaller and simpler particles, it is taken up through the process of endocytosis by the cells of the gastro-vascular cavity lining the intestine. Inside the cells of the gastro-vascular cavity, the remaining process of digestion takes place. Since it takes place inside the cell of the gastro-vascular cavity, it is now termed as intracellular digestion. So the process of digestion in planaria is both extracellular as well as intracellular. The digested food then diffuses through the cells of the gastrovascular cavity into the mesenchyme cells. The highly branching system of intestine actually helps in quick disposal of the food to all cells of the body, thus it also serves as alternate to the circulatory system. The undigested food in the intestine is egested through the mouth.

11.2.1 HUMAN DIGESTIVE SYSTEM

The digestive tract of human is **complete**. It is tube-like gut with two openings, mouth and anus. It is also termed as **gastro-intestinal tract** (G.I.T.). Like other animals with complete gut, it begins with mouth and terminates on anus. Human digestive system consists of digestive tract and accessory glands. Its various regions are specifically designed for different processes of holozoic nutrition. Associated with the G.I.T. are well elaborated accessory glands like salivary glands, liver, pancreas, etc.

The human digestive system consists of mouth, oral cavity (buccal cavity), pharynx, esophagus, stomach, small intestine, large intestine and anus. The glands associated with this are salivary glands, liver and pancreas. Let's discuss them one by one.

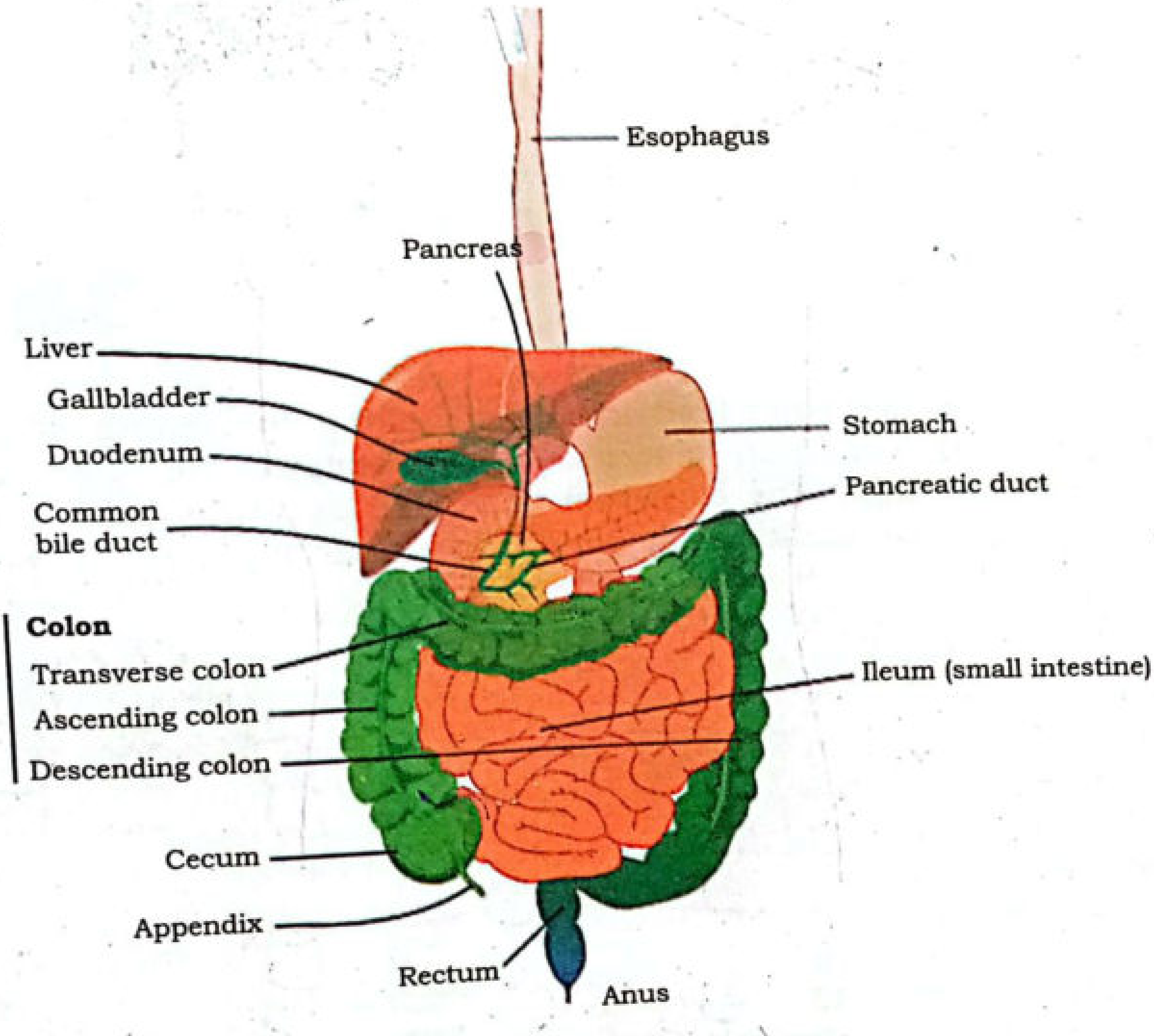


Fig: 11.3 Human digestive system

Mouth:

It is the anterior opening of alimentary canal bounded by two fleshy lips, termed as upper and lower lips, respectively. It is meant for ingestion of food.

Oral cavity:

The mouth opens into a wide space called oral cavity formed by upper and lower jaws. It has a muscular tongue on its floor. Its roof is formed by a dome-shaped bony, **hard palate** anteriorly and a **soft palate,** posteriorly. In the middle of the soft palate, hanging down is a muscular **uvula**. Laterally, it has cheeks, while it is connected posteriorly with a pharynx. Both jaws are lined by rows of teeth for cutting and grinding of food. They are meant for **mechanical digestion** in the oral cavity. The upper jaw is fixed while the lower is moveable.

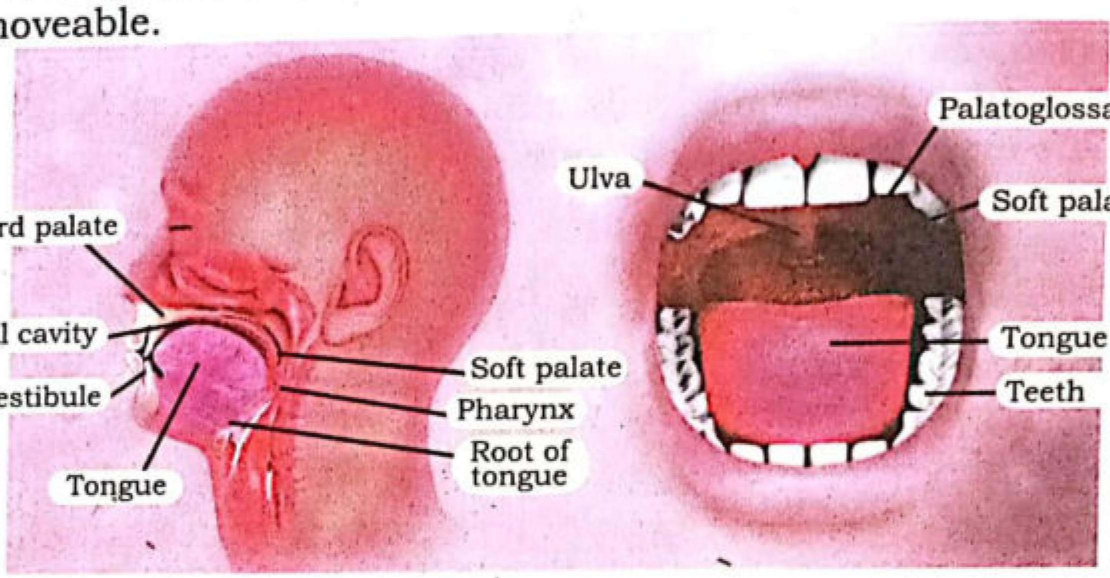


Fig: 11.4 Oral cavity

Teeth:

The teeth are embedded in sockets along the length of lower and upper jaws. This condition is termed as **thecodont**. We have two sets of teeth during life time. The condition is termed as **diphydont**. Initially, we have **deciduous or milk teeth** which are latter replaced by **permanent teeth**. The number of deciduous teeth is 20 while the number of permanent teeth is 32. Our teeth are different in shape and size, so the condition is termed as **heterodont**. This is correlated with different functions and different diet. Among the 32 permanent teeth, there are 8 Incisors, 4 canines, 8 premolars and 12 molars. The molars have no deciduous predecessors. The incisors are for cutting and

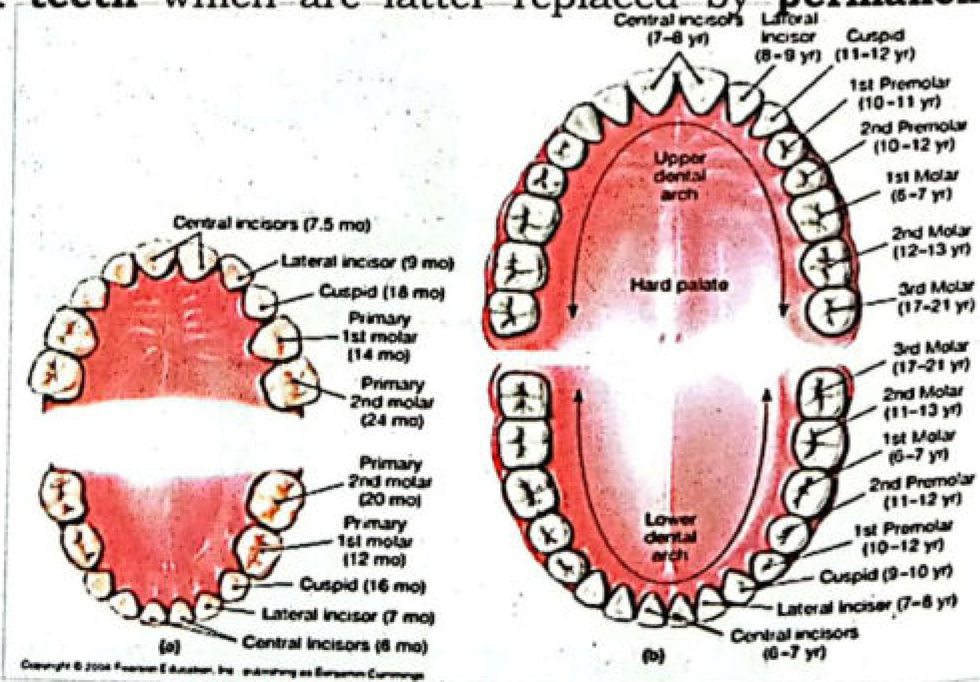


Fig: 11.5 Milk and permanent teeth

biting teeth and have flat sharp edges by which the food is cut into smaller pieces for ingestion. The sharp, pointed canines are for tearing and pulling flesh. The premolars and molars are used for grinding and crushing the food.

Tongue:

It's a muscular organ attached through hyoid bone with the floor of the oral cavity from its posterior and middle while free anteriorly at its tip. Its upper surface has numerous projections or **papillae** containing nerve endings for the sense of taste. The under surface of the tongue have a fold of mucous membrane called **frenulum**. The tongue helps in mastication, swallowing, taste and speech.

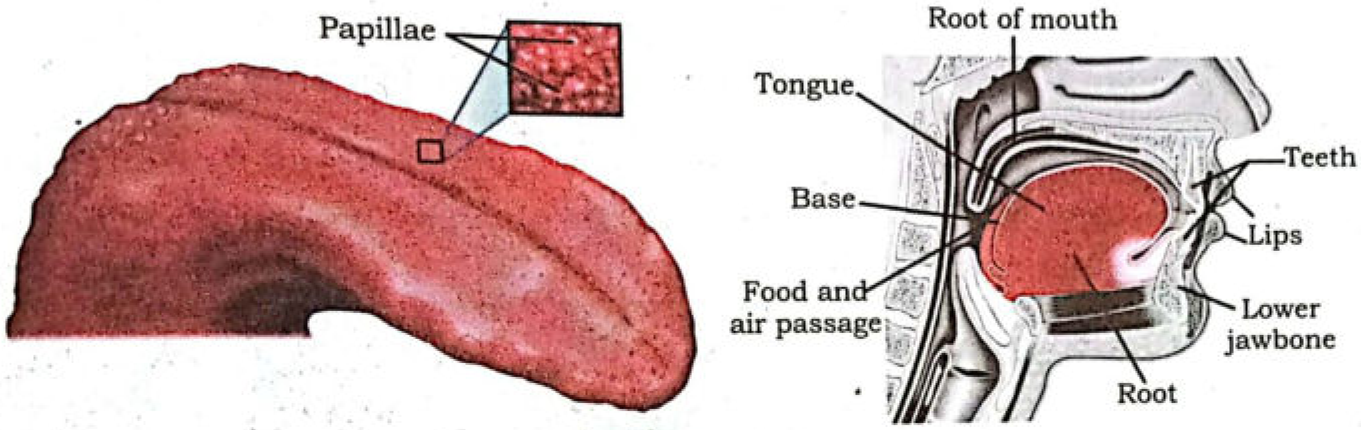


Fig: 11.6 Human tongue

Salivary Glands:

When you think about taking some spicy food, your mouth instantly get watered. This is actually a watery secretion, the **saliva**, secreted from the salivary glands present in your oral cavity. The saliva helps in lubrication of food through its **mucin (mucous)** which makes its swallowing easy. Its enzyme, **salivary amylase** also partially digests starch into maltose. It also helps in killing bacteria through its enzyme, **lysozyme**. Thus, the process of **chemical digestion** begins in the oral cavity. The human oral cavity has 3 pairs of major salivary glands, viz. **Parotid glands** at the base of the pinnae, **Sub-lingual glands** at the base of the tongue, and **Sub-mandibular glands** at the base of the lower jaw.

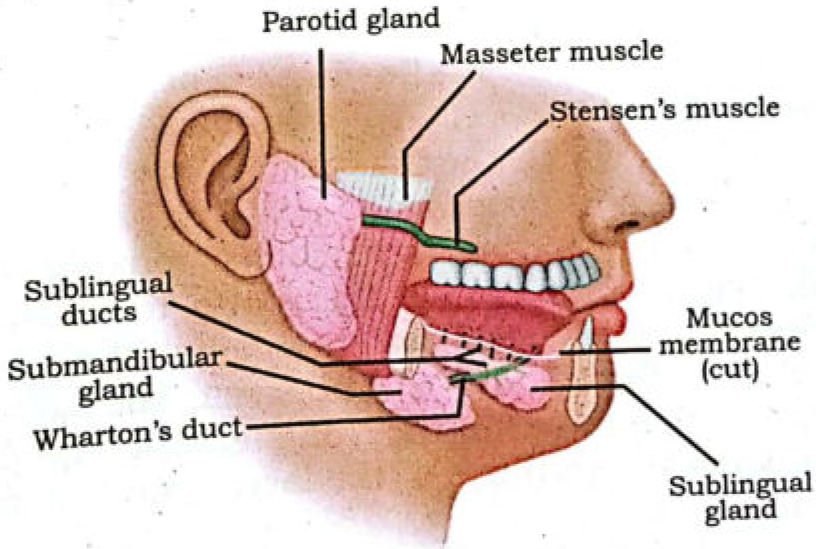


Fig: 11.7 Salivary glands

The salivary glands are controlled by autonomic nervous system. Normally, the amount of saliva is constant. It is reported that a normal person secretes 1 to 1.5 liters of saliva in 24 hours.

Pharynx:

The pharynx is connected anteriorly to oral cavity and nasal cavity simultaneously, while posteriorly connected to the esophagus and larynx. The pharynx permits the passage of swallowed solids and liquids food into the esophagus, or gullet, and conducts air to and from the trachea, or windpipe, during respiration. The pharynx also connects on either side with the cavity of the middle ear by way of the **Eustachian tube** and provides for equalization of air pressure on the eardrum membrane, which separates the cavity of the middle ear from the external ear canal. The principal muscles of the pharynx, involved in the mechanics of swallowing, are the three **pharyngeal constrictors**, which overlap with each other slightly and form the primary musculature of the side and rear pharyngeal walls.

Swallowing:

After sufficient chewing of the food and mixing up of saliva in the oral cavity, the food you ingested becomes a soft, lubricated ball-like structure called **bolus**. It is being pushed behind by the tongue into the pharynx. This process is termed as **swallowing or deglutition**. It is a voluntary process in which the mastication is ceased, air passage way is temporarily blocked, so bolus is pushed behind by the tongue into the pharynx. Entry of the bolus into the nasal pharynx is prevented by the elevation of the soft palate against the posterior pharyngeal wall. As the bolus is forced into the pharynx, the larynx moves upward and forward under the base of the tongue. The **epiglottis**, a lid-like covering that protects the entrance to the larynx, diverts the bolus to the esophagus.

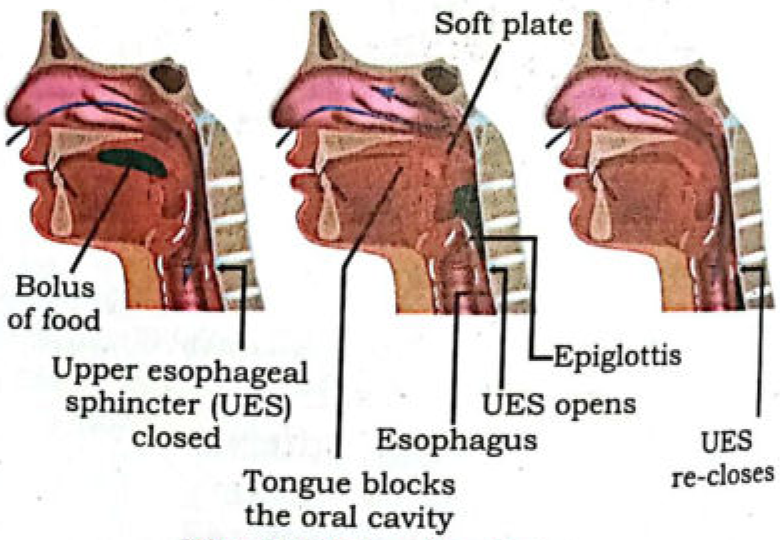


Fig: 11.8 Swallowing

Peristalsis:

A lot of times especially after taking large meal, you feel some movement in your abdomen. This is actually a wave like involuntary, rhythmic movement of contraction and relaxation of your gut muscles which propels the food forward inside the gut. This kind of movement also occurs in the tube-like or ducts of other systems of the body. It begins from the pharynx and then esophagus and continues propelling the food till the end of the digestive tract. The walls of alimentary canal contains smooth muscles controlled by autonomic nervous system. The muscular layer of the gut consists of inner layer of **circular muscles** while an outer layer of **longitudinal muscles**. The usual stimulus for peristalsis is distension of the gut wall due to bolus. The distension is due to the relaxation of circular while contraction of the longitudinal muscles. The circular muscles behind the bolus contract while the longitudinal muscles are relaxed simultaneously. This produces a propulsive force on the bolus.

Antiperistalsis:

Sometime due to over-eating, we feel some abnormal movement of the gut. This feeling is termed as **nausea**. We feel that the food in gut would come back through mouth and cause **vomiting**. It could be due to over-distension or excessive GIT irritation, a consequence of some poisonous food or toxic chemicals. It is reverse involuntary movement of smooth muscles unlike peristalsis. During the process, the stomach is squeezed. The gastro-pharyngeal sphincter is relaxed which causes the contents of the stomach to move upward through the esophagus in the form of vomiting.

Esophagus:

This tubular structure leads the bolus from pharynx to the stomach. It is located between trachea and spinal cord. Internally, it is lubricated by mucous which makes the passage of food through the esophagus easier. Through peristalsis the food is propelled through it into the stomach. The esophagus passes through the diaphragm before entering into the stomach. There is no process of mechanical as well as chemical digestions in esophagus.

Stomach:

It is a "J" shaped muscular, pouch-like structure which is associated with esophagus anteriorly and duodenum, posteriorly. It lies in the upper left of the human abdominal cavity. The top of stomach lies against the diaphragm. It receives food (bolus) from the esophagus. It's both ends are guarded by valves, anteriorly it has **cardiac sphincter** while posteriorly it has **pyloric sphincter**. The sphincters prevents the backward flow of gastric contents.

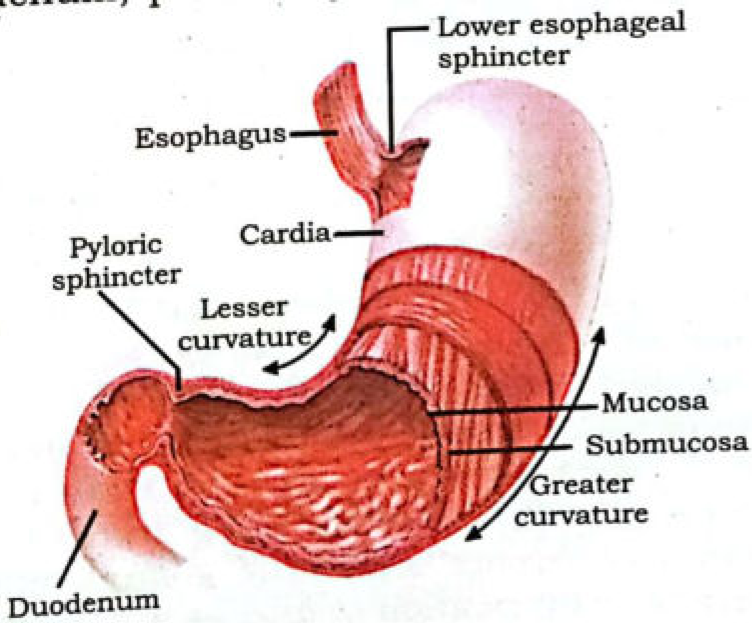


Fig: 11.9 Stomach

Stomach is the main organ of the digestive system where the processes of mechanical as well as chemical digestions take place. It can store food for several hours.

After several hours of digestion, it turns food into paste-like acidic, **chyme**.

The stomach is divided internally into four regions: the cardia, fundus, corpus (body) and pylorus. The esophagus opens into the cardia through cardiac sphincter. The fundus is the upper curved part which adapts to the varying volume of ingested food by relaxing its muscular wall. It usually contains a gas bubble, especially after a meal. The largest part of the stomach is the central, corpus (body) which primarily serves as the main reservoir of the ingested food. The lower most part of the stomach is pylorus which empties the gastric contents into the duodenum through pyloric sphincter.

The wall of the stomach consists of the usual four layers as present in other parts of the gastrointestinal tract. The inner most layer, **mucosa** is relatively thicker and contains numerous tubular glands. Under the mucosa lies the **sub-mucosa** containing blood vessels, nerves, etc. Below, sub-mucosa lies **mucularis externa.** It is also thick and, in some areas, it consists of 3 layers of smooth muscle, although this layering is not always visible. The three layers of smooth muscles are inner most oblique, the middle circular and the outer longitudinal layers. Glands are absent in the submucosa. The fourth and outermost layer is thin layer, the **serosa**. It secretes a lubricating fluid that reduces friction from muscle movement.

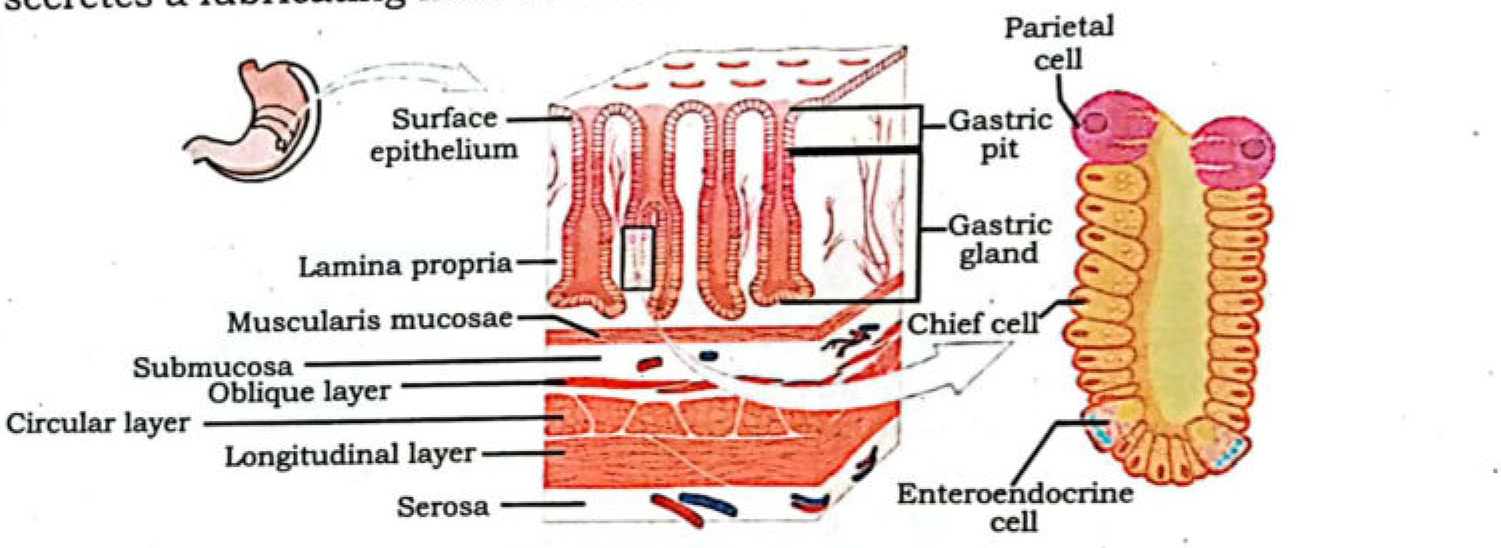


Fig: 11.10 T.S of stomach

In the empty contracted stomach, the mucosa is thrown into longitudinal folds or **rugae** because of the contraction of the muscularis mucosae and the loose consistency of the submucosa. The surface is further subdivided by furrows, the **gastric pits**. These funnel-shaped invaginations of the epithelium are continuous at their base with the tubular glands. The mucosa has mucous secreting **mucoid cells** throughout. It secretes mucous and bicarbonate ions. The gastric mucous is a glycoprotein which is a means of lubrication of food as well as protection to the stomach against the action of HCl and its own proteolytic enzyme. On the basis of differences in the types of glands present in the mucosa, three histological regions can be distinguished in the stomach. The first region around the cardia contains the **cardiac glands**. They secrete mucous only. The second region, which includes the fundus and corpus, contains the **gastric glands proper or fundic glands.** Here the main mucoid cells are lying. Besides the mucoid cells, there are **parietal cells or oxyntic cells** which secrete HCl (pH 1.5 to 2.5) and an **intrinsic factor.** The latter is important in the maturation of red blood cells, vitamin B12 absorption in intestine, and the health of certain cells in the central and peripheral nervous systems. The gastric glands also contain **Chief Cells or zymogen cells** which secrete **pepsinogen**, a precursor to proteolytic enzyme, **Pepsin**. The distal region of the stomach (pylorus) contains pyloric glands which secrete mucous and **gastrin** hormone. The gastrin is released in response to two basic stimuli, i.e., the distention of the stomach and the raising the pH of the stomach by the addition of foods like proteins (decrease in the acidic environment). In addition to the gastrin hormone, there are endocrine cells distributed throughout the body of the stomach. One of their important secretion is **serotonin** which inhibits the gastric acid secretions.

The physiological processes of mechanical and chemical digestions in stomach:

As soon as the food enters the stomach, both mechanical and chemical digestions start simultaneously. As a consequence of food anticipation (smell, taste, sight, sound), the stimulation signals are released by vagus nerve to the brain, or the distention of stomach due to actually food input especially proteins in stomach, the **gastrin** hormone is released into the blood. The vagus nerve represents the main component of **the parasympathetic nervous system**, which oversees a vast array of crucial bodily functions, including control of digestion, immune response, heart rate etc. The gastrin via blood stimulates the gastric glands to release their gastric juice into the lumen of the stomach. The gastric juice, as already pointed out, contains HCl and pepsinogen. The **HCl** softens the food, kills the germs, and converts the inactive **pepsinogen** into active proteolytic enzymes, the **pepsin**. The pepsin breaks down proteins into peptones (short chain polypeptides). The stomach itself is protected against the action of HCl through its thick coating of mucous. The pH of the mucosal layer is raised up to 7 due to the release of bicarbonate ions while the pH towards the lumen of the stomach remains around 1.5 to 2.5. Therefore, the factors affecting the gastric juice secretion are neuronal, mechanical and hormonal. In infants, another proteolytic enzyme, **rennin or chymosin** is secreted that turns soluble milk protein, caseinogen into insoluble protein, casein which is then digested by pepsin.

The process of mechanical digestion in stomach is the consequence of smooth muscles of the stomach which, as already pointed out are arranged in 3 layers, the inner most is the unique and obliquely orientated, the middle one is circular and the outer longitudinal smooth muscles. The contracting and peristaltic movements due to these muscles are responsible for the churning of the food in stomach, mixing up of the gastric juice with food and then transport of the food from stomach to duodenum. The food after several hours of mechanical and chemical digestion becomes a paste-like **chyme** which is then transported bit by bit to first segment of the small intestine, the duodenum. The process of mechanical digestion through its breaking down of food particles into smaller parts thus increases the surface area of the food for chemical activity of the digestive juices.

Absorptive role of the stomach:

Besides playing a major role in digestion of food, the stomach contributes a minor role in the absorption process also. It absorbs few of the digestive products, water, glucose, some other simple sugars, amino acids, and some fat soluble substances.

Small Intestine:

Stomach is followed by a long, coiled tube, the **small intestine**. It is about 6 meters long and 3 to 4 cm wide. It can be divided into three main regions, duodenum, jejunum and ilium. The small intestine is overall involved in completing the process of digestion and also the process of absorption.

Duodenum:

The first part is **duodenum** which is short in length. It is about 30 cm long and attached with pyloric stomach. At the junction of these two, there is valve, the **pyloric sphincter.** The duodenum forms a "C" shaped curve. Its initial part forms a bulb like dilation, the duodenal bulb. The duodenum receives a common bile duct and a pancreatic duct opening by a common aperture through which the secretions, bile and pancreatic juice are received from liver and pancreas, respectively. The bile is formed by the liver and stored in the gall bladder. The stimulation of gall bladder to release bile takes place by a peptide hormone of duodenum, the **cholecystokinin** which also plays role in pancreatic stimulation to secrete its pancreatic juice. Upon stimulation of gall bladder, bile is released into the **cystic duct** which is connected with the **common bile duct** before finally reaching the duodenum. The partly digested proteins of acidic chyme that entered the duodenum serve as major stimulus for cholecystokinin secretion. On the other hand, cholecystokinin also stimulates acinar cells of the pancreas to secrete large amount of pancreatic juice. Besides cholecystokinin, the cells of duodenum also secrete another hormone, **secretin** which acts upon pancreas to secrete water and bicarbonate ions. After that, the pancreatic juice is flushed out into the duodenum. The bicarbonate ions neutralize and turn the acidic chyme into alkaline.

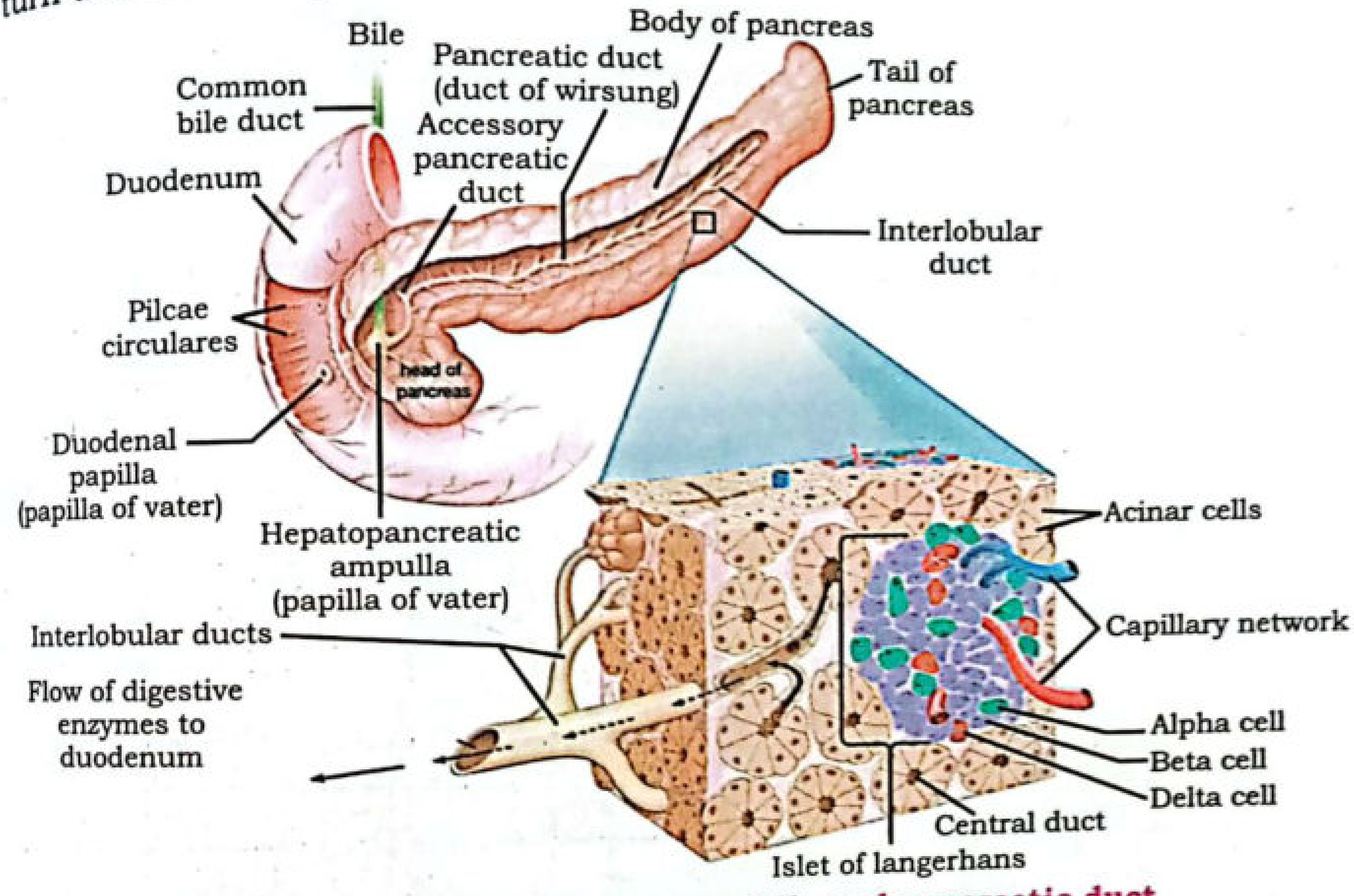


Fig: 11.11 Common bile and pancreatic duct

Bile is yellowish green in color and it contains water, salts, and bile pigments (brownish yellow **bilirubin** and greenish **biliverdin**). Its salts sodium glycholate and sodium taurocholate are involved in the emulsification of fats. There is no enzyme in bile. The bile pigments have nothing to do with the process of digestion. They are formed as excretory products of worn out (RBCs)

The pancreatic juice contains inactive enzyme **trypsinogen** which is converted into active **trypsin** by the action of **enterokinase** enzyme secreted by duodenum. The trypsin acts on proteins and peptones to break them into short chain polypeptides. Another proteolytic enzyme, **chymotrypsin** in the pancreatic juice converts casein into short chain amino acids. The other enzyme, the pancreatic acts upon starch and glycogen to break them into maltose (a disaccharide). **Lipase** of the pancreatic juice breaks down emulsified fats into fatty acids and glycerol. This completes the digestion of fats in the small intestine. The duodenum contains specific **Brunner's glands**, which produce a mucus-rich alkaline secretion containing bicarbonate ions. These secretions, in combination with bicarbonate from the pancreas, neutralize the stomach acids contained in gastric chyme.

Jejunum:

The partly digested food from the duodenum passes into the next region of the small intestine, the jejunum. It is the middle region of the small intestine and about 2.5 meter long. The region is specialized for digestion and absorption of the digested food. The process of digestion is completed by the different enzymes secreted by jejunum. It secretes maltase, sucrase, lactase and peptidase to digest maltose, sucrose, lactose and small peptides, respectively into their simple forms. Nucleotides are broken down into nucleosides by its nucleotidase. The cellulose remains undigested here.

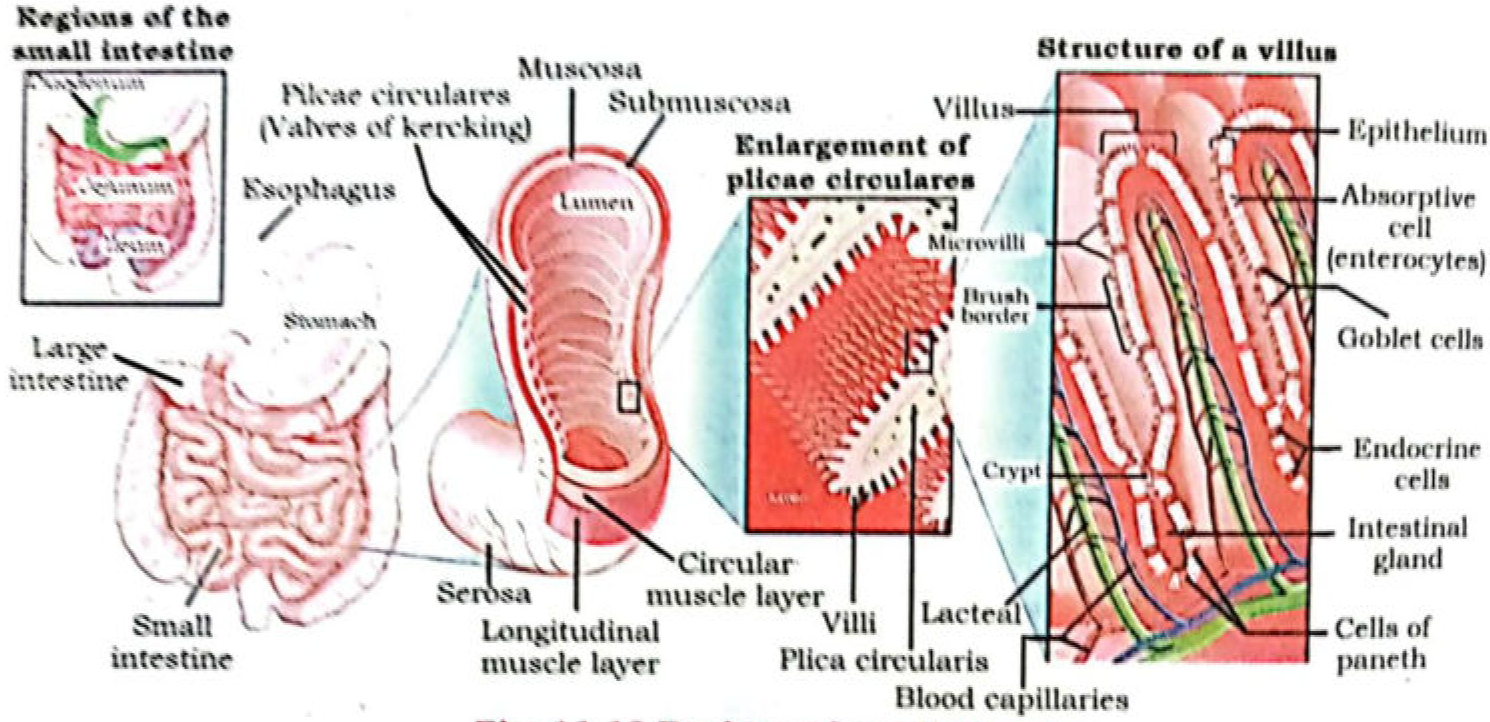


Fig: 11.12 Regions of small intestine

Ileum:

This is the last region of the small intestine. It is about 3.6 meters in length. The function of the ileum is mainly to absorb vitamin B12, bile salts, and any products of digestion that were not absorbed by the jejunum.

Internally, the walls of small intestine has wrinkle or folds called **plicae circulares.** From each of these folds arise numerous microscopic finger-like projections, **villi.** The individual epithelial cells of each villus membrane have microscopic hair-like projections, **microvilli**. Due to the villi, the ileum has an extremely large surface area for absorption. The plicae

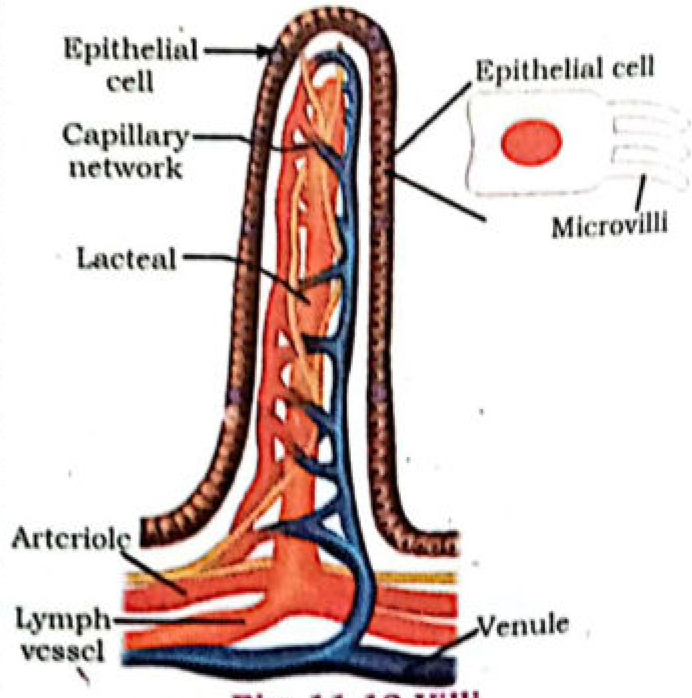


Fig: 11.13 Villi

circulares, villi and microvilli increase the surface are of the small intestine several folds for the absorption of soluble food. Each villus has a dense network of blood capillaries, blind lymph vessels called lacteals, and smooth muscles. The amino acids and monosaccharides are absorbed by the blood capillaries either by diffusion or active transport, while fatty acids and glycerol are taken up by lacteals. Smaller lymph vessels merge to form large vessels that drain fats (chylomicrons) and nutrients to the thoracic duct which is then emptied into subclavian vein. Some of the lipoprotein is taken up by the liver and exported into the blood stream. The blood capillaries containing fats fuse with each other to form **hepatic portal vein** which supplies all the nutrients to the liver for the storage, distribution and metabolism. Due to smooth muscles, the villi can contract and relax, constantly thus bringing themselves into close contact with food.

Large Intestine:

It is the last segment of the human digestive tract. The small intestine is connected with it. It is comparatively, shorter in length (about 1.5 meters) but greater in diameter (about 6.5 cm). The main functions performed here are reabsorption of water and nutrients, synthesis of some vitamins, reabsorption of some electrolytes such as sodium and chloride, formation of feces and then its egestion outside the body. It has mucous glands also for mucous secretion which helps in lubrication of the intestinal contents for its easy passage through the intestine. It lacks villi.

Large intestine is divisible into following regions, viz., caecum, colon, rectum and anal canal.

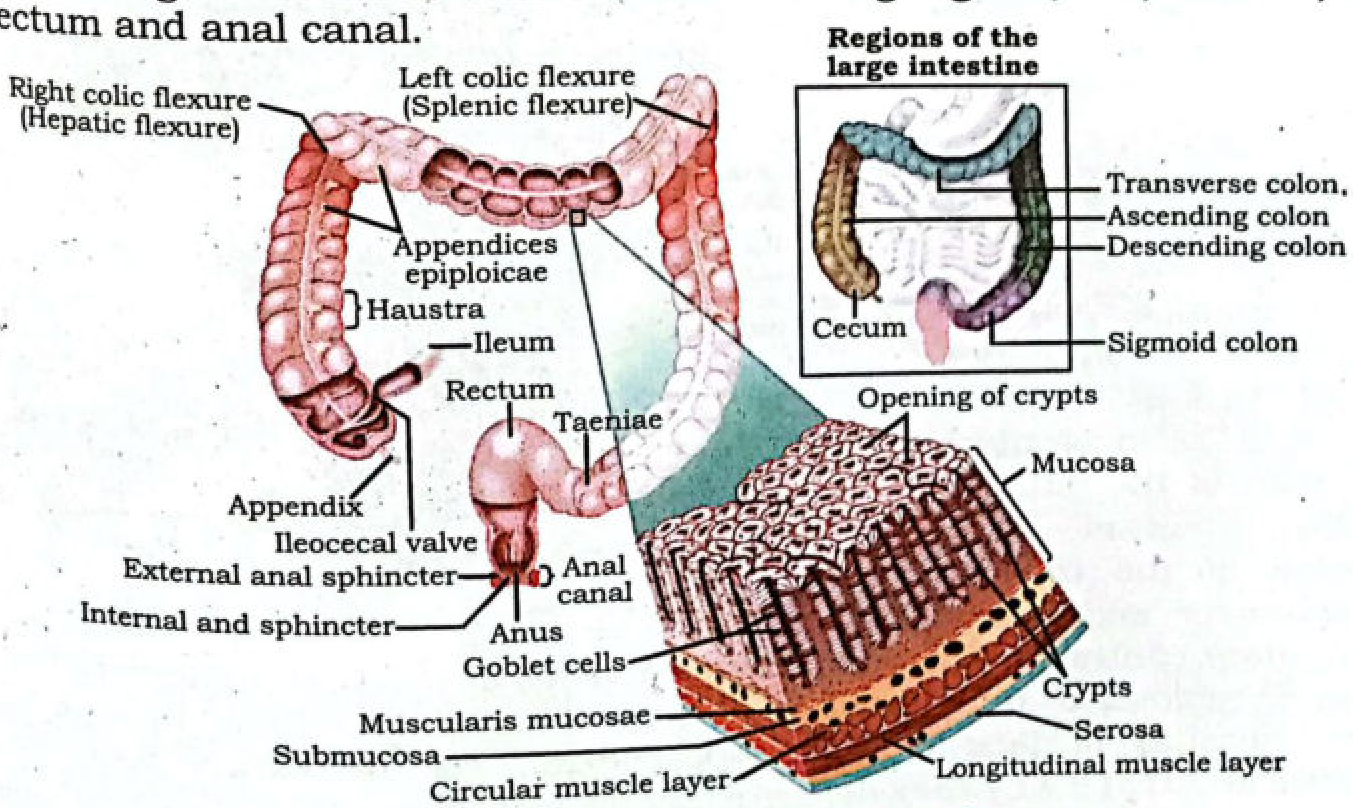


Fig: 11.14 Regions of large intestine

Caecum:

Small intestine joins the large intestine at a short **caecum**. It is located in the lower right side of the abdominal cavity. At the junction of small and large intestines, there is an **ileocecal valve**. It controls the passage of undigested food from ilium to the caecum. Water and salts absorption takes place here. The caecum, from its lower side, gives out a blind tube of about 18 cm long called **vermiform appendix**. It is a vestigial organ.

Colon:

The next region after caecum is colon. It consists of ascending colon, transverse colon, descending colon and sigmoid colon. It is involved in the reabsorption of water, salts and vitamins. The colon also contains large numbers of symbiotic bacteria (e.g., Lactobacillus, etc.) that synthesize niacin (nicotinic acid), thiamin (vitamin B1) and vitamin K, vitamins that are essential to several metabolic activities as well as to the function of the central nervous system.

Rectum:

The sigmoid colon opens into rectum. It is about 13 cm long and terminate at anal canal. The rectum ends at dilated portion, **the rectal ampulla.** When the rectum is filled with feces, the impulse to defecate occurs. The anal canal has an internal sphincter valve of smooth muscles and an external sphincter valve of skeletal muscles. In adults, the defecation involves both the involuntary contraction of the muscles of the rectum and relaxation of the internal anal sphincter and finally the voluntary contraction of the skeletal muscles of external sphincter. The process of removal of undigested food is egestion or defecation. In case of infants, because of lack of voluntary control, the defecation reflex is automatic and that's why they defecate inconveniently anytime.

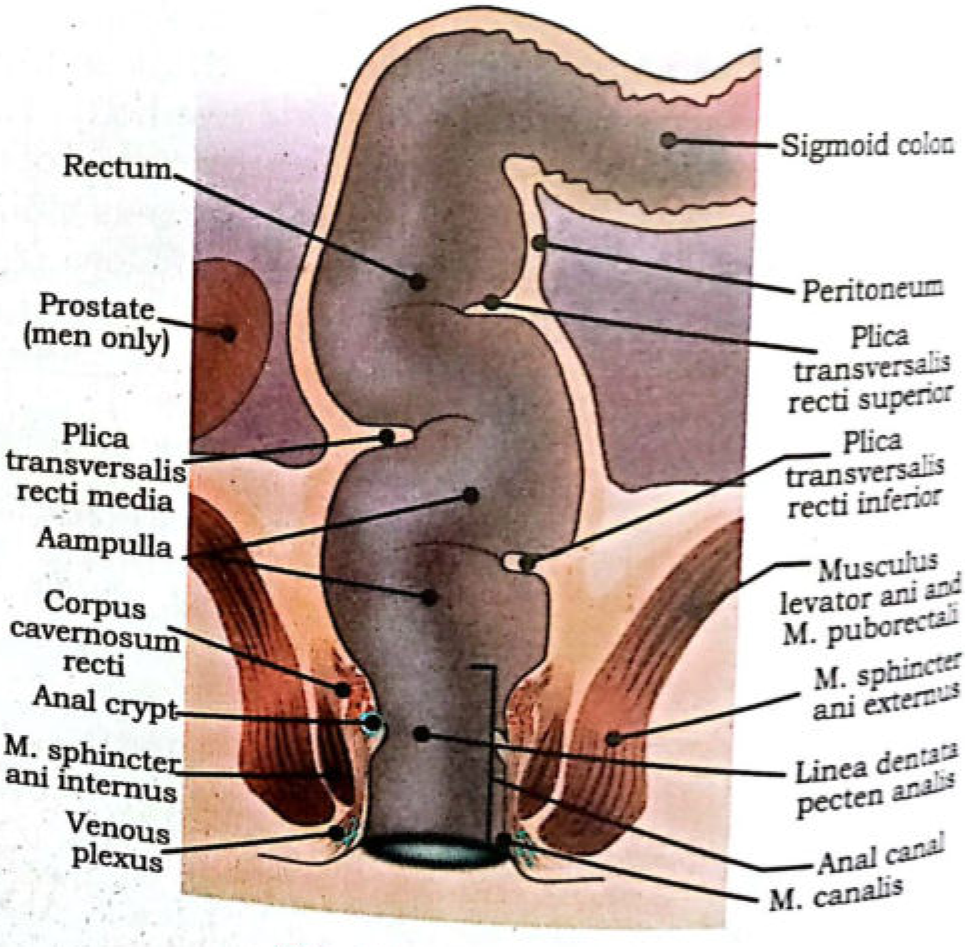


Fig: 11.15 Rectum

11.3 ROLE OF ACCESSORY GLANDS:

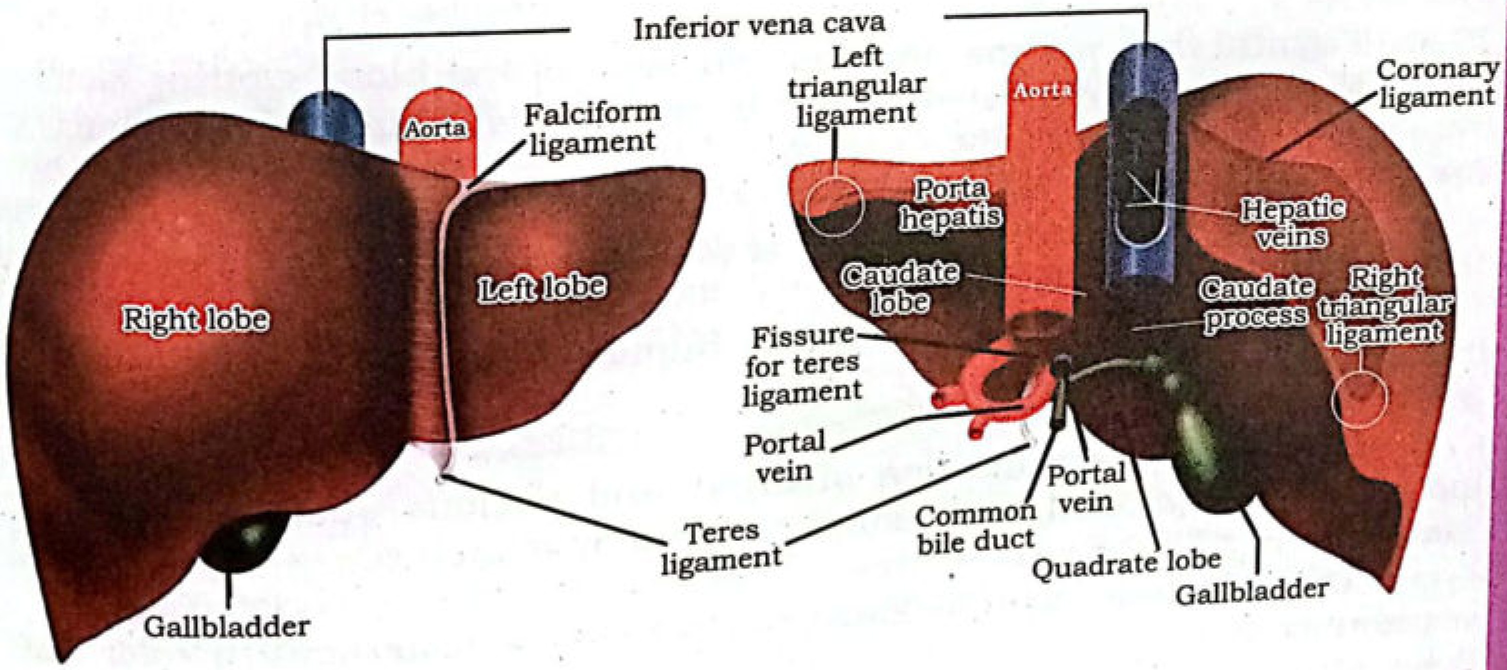
The accessory glands are those exocrine glands which upon appropriate stimulation secrete their secretions (juices) into the alimentary canal through specific ducts. These secretions help in the process of digestion. The accessory glands consist of 3 pairs of salivary glands, liver and pancreas.

Salivary glands:

The location, stimulation, secretion and the role of salivary glands have been already discussed earlier.

Liver:

The liver is the largest gland in our body. It is situated in the upper part of the abdominal cavity just behind the diaphragm more towards right side. The liver is enclosed in a thin inelastic capsule and incompletely covered by a layer of **peritoneum**. Folds of peritoneum form supporting ligaments attaching the liver to the inferior surface of the diaphragm. The liver has four lobes. The two most obvious are the large right lobe and the smaller, wedge-shaped, left lobe. The other two, the caudate and quadrate lobes, are areas on the posterior surface. The lobes of the liver are made up of tiny lobules just visible to the naked eye. These lobules are hexagonal in outline and are formed by cubical-shaped cells, the hepatocytes, arranged in pairs of columns radiating from a central vein.



Blood Supply to Liver:

The liver receives blood supply from two sources, one is hepatic artery which supplies oxygenated blood while the other one is hepatic portal vein which brings nutrient rich blood from various regions of the alimentary canal.

Functions of Liver:

Liver is a metabolic factory, synthesis and storage organs as well as homeostatic and secretory organ of our body. It performs hundreds of functions; among them, the important are as follows:

Secretion of bile:

The bile is synthesized by the hepatocytes. The bile consists of bile salts, bile pigments, and cholesterol.

Metabolism& Homeostasis:

It converts glucose into glycogen in the presence of the hormone insulin secreted by pancreas, and reconverts liver glycogen back to glucose in the presence of another hormone, glucagon, which is also secreted by pancreas. These changes are important regulators of the blood glucose level. It converts stored fat to a form in which it can be used by the tissues to provide energy. It is involved in deamination of amino acids thereby removing the nitrogenous portion from the amino acids not required for the formation of new protein; urea is formed from this nitrogenous portion which is excreted in the urine. It also breaks down the genetic material of worn-out cells of the body to form uric acid which is excreted in the urine. It removes the nitrogenous portion of amino acids and attaches it to other carbohydrate molecules forming new non-essential amino acids.

Synthesis:

It synthesizes plasma proteins and most of the blood clotting factors from the available amino acids occur in the liver. It synthesizes vitamin A from carotene.

Breakdown:

It is involved in breaking down of worn out erythrocytes.

Defense:

This is carried out by phagocytic Kupffer cells (hepatic macrophages) in the sinusoids.

Detoxification:

It performs detoxification of drugs and noxious substances. These include ethanol (alcohol) and toxins produced by microbes.

Inactivation of hormones:

These include insulin, glucagon, cortisol, aldosterone, thyroid, and sex hormones.

Production of heat:

It uses a considerable amount of energy, has a high metabolic rate, and produces a great deal of heat. It is the main heat-producing organ of the body.

Storage:

It stores fat-soluble vitamins: A, D, E, K; iron, copper; some water-soluble vitamins, e.g. riboflavin, niacin,pyridoxine, folic acid and vitamin B12.

Some diseases related to Liver:

Hepatitis:

It refers to the inflammation of liver. There could be several factors responsible for hepatitis such as autoimmune disorder, viral infection, some toxins, drugs, or drinking alcohol. Hepatitis is commonly characterized by fatigue, flu-like symptoms, pale skin, abdominal pain, dark urine, and pale stool. Infectious hepatitis is accompanied by high fever.

Jaundice:

It is a yellowish discoloration of the skin, mucous membranes and of the white of the eyes caused by elevated levels of bilirubin in the blood (hyperbilirubinemia). Bilirubin is a yellow pigment that is created by the breakdown of dead red blood cells in the liver. Normally, the liver gets rid of bilirubin along with old red blood cells. Jaundice may indicate a serious problem with the function of your red blood cells, liver, gallbladder, or pancreas. Jaundice itself is not a disease, but it is a symptom of several possible

Pancreas:

It lies behind the stomach in horizontal line along the curve of duodenum. It is 12 cm to 15 cm long. Though it serves as both exocrine as well as endocrine gland, here we will discuss its exocrine role only. It consists of a large number of lobules, the walls of which consist of secretory cells. Each lobule is drained by a tiny duct and these unite eventually to form the pancreatic duct, which extends the whole length of the gland and opens into the duodenum. Just before entering the duodenum, the pancreatic duct joins the common bile duct to form the **hepatopancreatic ampulla**. The duodenal opening of the ampulla is controlled by the hepatopancreatic sphincter (of Oddi).

As an exocrine gland, the pancreas secretes pancreatic juice containing enzymes that digest carbohydrates, proteins, and fats (discussed earlier in detail).

11.4 DISORDERS RELATED TO DIGESTIVE SYSTEM AND FOOD HABITS:

Ulcer:

An ulcer is a sore (a painful wound) which could be developed anywhere in the body. In the gastro-intestinal tract, the most common ulcers are "peptic ulcers". They may be termed as gastric ulcers or intestinal ulcers depending upon their location in the stomach or intestine, respectively. They are also termed as "peptic ulcers" since the damage is caused by the digestive juices. It happens when the HCl of the gastric juice etches away the protective layer of mucous which may cause painful condition. The common causes of peptic ulcer aresome bacteria, like

Helicobacter pylori (H. pylori), some medication like aspirin, stress, drinking alcohol or taking lots of spicy food. The symptoms of ulcers includes, loss of appetite, sharp abdominal pain, nausea, and vomiting. The treatment depending upon the causes consists of proper antibiotics and proton-pump Inhibitors (PPIs) to reduce the secretion of acid in stomach.

Food Poisoning:

Food poisoning or illness is caused by taking contaminated, spoilt or toxic food. The food may be contaminated through some bacteria such as *Staphylococcus*, some viruses, parasites, or other pathogens. The uncooked food is a common factor for food contamination. The common symptoms of food poisoning are stomach cramps, vomiting, fever and diarrhea. The treatment is related to rehydration, electrolyte solutions and some antibiotics.

Dyspepsia:

It is popularly known as indigestion. It refers to the discomfort or pain that occurs in the upper abdomen followed by eating or drinking. The common symptoms includes blotting, nausea and abdominal fullness. It may cause heart-burn due to acid reflux.

Obesity:

It is related to the accumulation of excessive body fats. Medically, according to WHO, at adult age (around 35 years) person is considered obese when his body-mass Index (BMI) is over 30 kg/m^2. BMI is obtained by dividing the weight of a person with the square of his height. Obesity is the result of combination of inheritance, environment, diet and exercise. The obesity increases the risk of diabetes, cardiovascular diseases, hypertension and some cancers. Even though medications and diets can help, the treatment of obesity cannot be a short-term "fix" but has to be a lifelong commitment to proper diet habits, increased physical activity, and regular exercise.

Anorexia nervosa:

It is a psychological disorder in which the person has fear of gaining weight so refuse to eat appropriately. It is commonly associated with young girls and women. As a result, the victim develops anorexia (loss of appetite), the weight reduces to normal. It may be accompanied by spontaneous or induced vomiting in some patients.

Bulimia nervosa:

Just like anorexia nervosa, it is also psychological disorder of gaining excessive body weight. This may be a very serious, life threatening disorder. It is generally characterized by binge eating (episodes of uncontrollable eating so the person consumes excessive amount of food) followed by purging (induced vomiting). Complications from bulimia may lead to kidney failure, heart problems, teeth decay, electrolyte or chemical imbalances.

SUMMARY

- The nutrients are required by protoplasm to perform its biological functions.
- Holozoic nutrition is one of the type heterotrophic nutrition occurs in animals.
- Holozoic nutrition is consist of ingestion, digestion, absorption, assimilation and egestion.
- Ingestion is the taking of food into the cell
- Digestion is the process of breaking down of complex or non-diffuseable food into simple or diffudeable molecules.
- Absorption- soluble food molecules are absorbed in the cell or transporting fluid.
- Assimilation- utilization of absorbed food molecules in the cell during metabolic activities.
- Assimilation- utilization of absorbed food molecule in the cell during metabolic activities.
- Egerton- Removal of undigested food.
- There are wo types of digestion, intercellular and extracellular, another way chemical or mechanical digestion.
- Digestion occurs in sac like or tube like digestive tract.
- The digestive tract of human consist of mouth, oral cavity, pharynx, esophagus, stomach, small intestine, large intestine and anus.
- Mechanical digestion takes place in oral cavity and stomach.
- Chemical digestion occur in oral cavity, stomach, duodenum and jejunum.
- The human digestive system also contain salivary glands, liver, gall bladder and pancreas and some glands present in stomach and jejunum.
- Hepatic portal vein bring nutrient rich blood from various regions of alimentary canal to liver.
- Liver is a metabolic factory, synthesis and storage organ as well as homeostatic and secondary organ.
- Pancreas serves as both exocrine and endocrine gland.
- Main disorder of digestive system are ulcer, food poisoning, dyspepsia, obesity, anorexia and bulimia nervosa.

EXERCISE

1. Encircle the correct choice.

(i) The process of nutrition does not include this:
- (a) Taking in of food (b) Breaking down of food
- (c) Absorption of nutrients (d) Removal of undigested food

(ii) The chemicals playing the most important role in digestion of food are:
- (a) Hormones (b) Enzymes
- (c) HCl (d) Both a & b
- (e) None of these

(iii) The thecodont condition refers to:
- (a) Presence of two sets of teeth during life time
- (b) all alike teeth
- (c) Sharp, pointed teeth
- (d) Embedded teeth

(iv) The type of digestion in Planaria is:
- (a) Intracellular (b) Extracellular
- (c) Intercellular (d) Both a & b

(v) Number of milk teeth is:
- (a) 10 (b) 15
- (c) 20 (d) 25

(vi) Teeth meant for cutting are:
- (a) Incisors (b) Canines
- (c) Pre-molars (d) Molars

(vii) The part of gut also associated with respiratory system is:
- (a) Stomach (b) Esophagus
- (c) Pharynx (d) Both b & c

(viii) The number of salivary glands in our oral cavity are:
- (a) 3 (b) 4
- (c) 5 (d) 6

(ix) The rugae are related to:
- (a) Stomach (b) Duodenum
- (c) Jejunum (d) Ilium

(x) BMI over 30 kg/m^2 is considered as:
- (a) Normal (b) Over-weight
- (c) under-weight (d) Obesity

2. Write short answers of the following questions:

1. List out the stages in holozoic nutrition.
2. How stomach itself is protected against the strong HCl?
3. Point out the ways through which the internal surface area of the small intestine is increased.
4. How bile help in the digestion of fats?
5. Enlist the role of large intestine.
6. What accounts for the presence of bacteria in large intestine?
7. What are the health risks involved in obesity?
8. List out some factors which may lead to obesity.
9. What is Anorexia nervosa?

3. Write detailed answers of the following questions:

1. State and explain the process of digestion in Amoeba?
2. Categorize the animals based upon their size of the food? Give examples of each group.
3. Discuss the process of digestion in stomach?
4. State and explain the role of accessory glands associated with our gut.
5. Explain the process of digestion and absorption of food in small intestine.
6. State and explain the sac-like gut with suitable example.